PRODUCTS & COMPATIBILITY

Product Specifications

Product specifications describe the technical characteristics, capabilities, and limitations of hardware sold by PixelForge. Understanding these specifications helps customers compare products and select components that meet their requirements.

Specifications are provided using information supplied by manufacturers and may include details about performance, dimensions, compatibility, power requirements, and supported technologies.

Understanding Processor Specifications

Processors are commonly described using their model name, architecture, core count, thread count, clock frequencies, cache capacity, and supported memory standards.

Core count indicates the number of physical processing cores available. Thread count describes the number of execution threads exposed by the processor, which may differ from the number of physical cores.

Clock frequency is usually measured in gigahertz. Higher clock frequencies can improve performance in certain workloads, but they should not be used as the only measurement when comparing processors from different architectures.

Cache capacity, power limits, and processor architecture can also significantly affect real-world performance.

Customers should verify socket compatibility and motherboard BIOS support before purchasing a processor.

Understanding Graphics Card Specifications

Graphics cards are commonly compared using their graphics processor, video memory capacity, memory interface, supported technologies, and power requirements.

Video memory, often called VRAM, stores textures, frame buffers, and other graphical data used during rendering.

A graphics card with more VRAM is not automatically faster than one with less memory. Overall performance depends on the graphics processor, memory bandwidth, architecture, and software optimization.

Customers should also check the card's physical dimensions, cooling design, and required power connectors.

Performance results may vary significantly depending on the game, resolution, graphical settings, and processor used in the test system.

Monitor Specifications

Gaming monitors are commonly described by their screen size, resolution, refresh rate, response time, panel technology, and supported connections.

Resolution determines the number of pixels displayed. Common gaming resolutions include 1920 × 1080, 2560 × 1440, and 3840 × 2160.

Refresh rate is measured in hertz and describes how many times the display can update its image each second. A 144 Hz monitor can refresh up to 144 times per second, provided the computer and connection can supply the required signal.

Response time describes how quickly pixels change between states, but measurement methods vary between manufacturers. Advertised response times should therefore be interpreted carefully.

Adaptive synchronization technologies may help reduce visible tearing and stuttering when the monitor and graphics hardware support compatible features.

Memory Specifications

Computer memory is commonly described by its capacity, generation, transfer rate, latency, and module configuration.

For example, a 32 GB DDR5 memory kit may contain two 16 GB modules intended for installation in compatible motherboard slots.

Advertised memory speeds may require enabling XMP or EXPO settings. The actual supported speed depends on the processor, motherboard, firmware, and memory configuration.

Customers should confirm that the motherboard supports the memory generation and desired capacity before purchasing.

Storage Specifications

Solid-state drives are commonly described by their storage capacity, interface, form factor, and sequential read and write speeds.

SATA SSDs and NVMe SSDs use different communication technologies. NVMe drives connected through PCI Express can offer substantially higher transfer rates than traditional SATA SSDs.

Sequential transfer speeds describe performance when reading or writing large continuous files. Everyday responsiveness also depends on random access performance, workload characteristics, and system configuration.

A faster advertised sequential speed does not guarantee an equally large improvement in game loading times or ordinary application use.

Customers should verify the supported storage interface and physical dimensions before purchasing a drive.

Keyboard and Mouse Specifications

Gaming keyboards may use mechanical, optical, or membrane-based switches. Switch characteristics influence key feel, activation behavior, and typing experience.

Keyboard specifications may also describe layout, connectivity, polling rate, lighting, and programmable features.

Gaming mice are commonly described by their sensor, sensitivity range, polling rate, weight, and connection method.

Sensitivity is often expressed in DPI or CPI. Higher maximum sensitivity does not automatically indicate better tracking accuracy or overall mouse quality.

Customers should consider ergonomics, intended use, and personal preferences alongside technical specifications.

Headset Specifications

Gaming headset specifications may include driver size, frequency response, microphone characteristics, connection type, and supported audio features.

A wider advertised frequency range does not automatically guarantee better sound quality. Acoustic design, tuning, microphone processing, and user preferences all influence the listening experience.

Wireless headsets may use Bluetooth or dedicated wireless receivers. Compatibility and latency can vary depending on the connection technology and device.

Product Images and Descriptions

Product images are intended to illustrate the appearance and general design of an item. Manufacturers may provide promotional images showing optional accessories or configurations that are not included in the standard package.

Customers should rely on the written product description and included accessories list to determine what is supplied.

If an important specification is unclear, customers should contact PixelForge support before placing an order.

Comparing Products

When comparing hardware, customers should consider the intended application, compatibility requirements, performance measurements, power consumption, and overall cost.

Manufacturer specifications provide useful technical information, but real-world performance may differ depending on software, system configuration, environmental conditions, and usage patterns.

Independent testing can provide additional context, particularly when comparing products with similar specifications.

For compatibility questions, see PC Components Compatibility. For complete gaming PC configurations and assembly procedures, consult Custom PC Builds.